Department of Physics

PCS 125–Midterm-solutions-Waves and Fields

MULTIPLE CHOICE QUESTIONS

1. A mass on a spring has an angular oscillation frequency of 2.56 rad/s. The spring constant is 27.2 N/m, and the system's kinetic energy is 4.16 J when t = 1.56 s. What is the oscillation amplitude? Assume that the mass is at its equilibrium position when t = 0.
a. 63.1 cm
b. 47.7 cm
c. 73.4 cm
d. 55.3 cm
e. 84.0 cm

2. A certain spring elongates 9:0mm when it is suspended vertically and a block of mass M is hung on it. If driven by a force of variable frequency, at what frequency would resonance be obtained?
a. is 0:088 rad/s
b. is 200 rad/s
c. is 33 rad/s
d. is 1140 rad/s
e. cannot be computed unless the value of M is given

3. A simple pendulum is suspended from the ceiling of an elevator. The elevator is accelerating upwards with acceleration a. The period of this pendulum, in terms of its length L, g, and a is:
a. $2\pi(L/g)^{1/2}$
b. $2\pi[L/(g+a)]^{1/2}$
c. $2\pi[L/(g-a)]^{1/2}$
d. $2\pi(g+a/L)^{1/2}$
e. $(1/2\pi)(g/L)^{1/2}$

4.For a wave described by the equation $y = 0.15\sin\left[\frac{\pi}{16}(2x - 64t)\right]$ (with all variables in SI units). Determine the first positive x-coordinate for which y is at its maximum value when t = 0.
a.16 m
b. 8 m
c. 4 m
d. 2 m
e. 13 m

5. A pulsed wave is described by the equation

$$y(x,t) = \frac{2}{(x-3t)^2 + 2}$$

where x and y are measured in cm and t in s. What is the speed of the wave and its direction?
a. 2.0 m/s in the negative x direction.
b. 2.0 m/s in the positive x direction.
c. 3.0 m/s in the negative x direction
d. 3.0 m/s in the positive x direction.
e. None of the above

6. What is the equation of a wave, traveling along the $+x$ axis with an amplitude of 0.02 m, a frequency of 440 Hz, and a speed of 330 m/s.

a. $y = 0.02 \sin [880\pi (x/330 - t)]$
b. $y = 0.02 \cos [880\pi x/330 - 440t]$
c. $y = 0.02 \sin [880\pi (x/330 + t)]$
d. $y = 0.02 \sin [2\pi (x/330 + 440t)]$
e. $y = 0.02 \cos [2\pi (x/330 + 440t)]$

7. A string with a fixed frequency vibrator at one end forms a standing wave with 4 antinodes when under tension T_1. When the tension is slowly increased the standing wave disappears until tension T_2 is reached, there being no resonances occurring between these two tensions. How many antinodes are there in this new resonance?
a. 2
b. 3
c. 4
d. 5
e. 6

8. An observer stands 3 m from speaker A and 5 m from speaker B. Both speakers, oscillating in phase, produce 250 Hz waves. The speed of sound in air is 340 m/s. What is the phase difference between the waves from A and B at the observer's location?
a. 0.273 rad
b. 1.81 rad
c. 3.32 rad
d. 2.96 rad
e. 4.27 rad

9. Two harmonic waves travelling in opposite directions interfere to produce a standing wave described by $y = 2\sin(4x)\cos(3t)$ where x and y are in m and t is in s. What is the transverse speed of the rope at $t = 3$s and $x = 1$ m?
a. 0.75 m/s
b. 1.9 m/s
c. 1.3 m/s
d. 12 m/s
e. 3.0 m/s

10.Two lead spheres of mass M are separated by a distance R. They are isolated in space with no other masses nearby. The magnitude of the gravitational force experienced by each mass is F. Now one of the masses is doubled, the other is tripled, and they are pushed farther apart so that their separation is now 3R. The magnitude of the gravitational force on the larger mass is now:
a. 3F
b. 2F
c. F/2
d. 2F/3
e. 3F/2

11.The period of a satellite circling planet Klaatu is observed to be 84 s when it is in a circular orbit with a radius of 8.0×10^{6} m. What is the mass of planet Klaatu?
a. 6.2×10^{28} kg
b. 5.0×10^{28} kg
c. 5.5×10^{28} kg
d. 4.3×10^{28} kg
e. 3.7×10^{28} kg

12.Two planets of radii R_1 and R_2 are made of the same material such that they have the same density (mass per unit volume). The ratio of their accelerations due to gravity (g_1/g_2) at their respective surface is:
a. $(R_1/R_2)^2$
b. R_1/R_2
c. $(R_2/R_1)^2$
d. R_2/R_1
e. 1

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